

TheFatCat Whitepaper

Decentralized Trading-Tax Routing Protocol with Adaptive Damping and Seniority Weighting

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TheFatCat Protocol Core Contributors · Protocol Rules and Calculation Models
This document contains no representations of anticipated return on investment or yield commitments

Abstract

Common decentralized dividend and trading-tax designs enforce globally shared reward assets, pass fee flows directly through with little temporal damping, and allocate proceeds by capital balance alone without seniority weighting. These are characteristics of upstream templates compared here, not universal claims across all fee-routing tokens.

TheFatCat composes the following mechanisms into an integrated protocol architecture:

1. **The Belly Hydrodynamic Reservoir:** Permissionless clock progression every 8 hours minimum. At nominal cadence, it disburses $\frac{1}{21} \approx 4.7619\%$ of unreserved quote balance per interval, yielding a zero-inflow model half-life of 4.736 days;
2. **O(1) Closed-Form Seniority Ledger:** Decomposes cohorts algebraically to aggregate seniority notches 1 through 22 without iterating over positions. Reward claiming scales with batches traversed; callers may set a smaller `claimThrough` cutoff;
3. **Sovereign Diet Architecture:** Autonomous asset selection per position. Ignoring observation limits and integer dust, nominal quote budgets depend on global weight share; asset choice affects execution, market value, issuer risk, and external incentives;
4. **Dual-Liability Accounting & Conservative Rounding:** WBNB default and fallback quote asset, batch accounting, verified delivery balance assertions, and integer floor truncation maintaining tested solvency bounds.

1 Introduction & Paradigm Shift

1.1 Structural Dilemmas in Direct Pass-Through Models

In decentralized automated market maker (AMM) ecosystems, trading-tax tokens typically route swap fees directly into uniform reward tokens for token holders. Upstream templates exhibit three structural constraints:

1. **Globally Shared Reward Selection:** All eligible holders share a single asset or fixed basket without per-position preference;
2. **Absence of Temporal Buffering:** Fee inflows drop sharply when trading volume contracts, transmitting volatility instantly;
3. **Absence of Seniority Input:** Allocation depends strictly on snapshot capital without rewarding duration. Transient capital can dilute historical holders during high-volume spikes.

1.2 The Reservoir Model: Damping Flow into Sticky Inventory

TheFatCat transforms the pass-through pipe into a self-damped reservoir, **The Belly reward treasury**. FATCAT launches paired with QQQB. After Flap deducts its platform fee, QQQB tax revenue enters `FatCatQqqbVault`, follows `QQQB → USDT → WBNB`, and is unwrapped for the BNB receiver. Its `flush()` allocates **85% of the remaining net revenue to the treasury** and **15% to the Ops Safe for project development, operations and marketing**. The exact shares are 17/20 and 3/20; the team amount is rounded down and every remainder goes to the treasury. Revenue splitting occurs only in the receiver.

The 4% FATCAT trading tax includes Flap's platform fee of **0.2% of eligible trade value**, equal to 5% of collected tax. Allocation percentages use the following calculation bases:

Allocation	Share	Calculation basis
Flap platform fee	0.2%	Eligible FATCAT trade value
Eligible FEED reward	0.1% target	Trade value represented by this conversion
Reward treasury	85%	Remaining protocol net revenue
Project development, operations and marketing	15%	The same remaining net revenue

The FEED reward targets 0.1% of the eligible trade value represented by this conversion. With a 4% tax and Flap retaining 5% of that tax, the protocol receives a nominal 3.8% of trade value. The reward is one thirty-eighth of actual WBNB conversion output (about 2.63158%), rounded down. The 0.1% target assumes unchanged value and no conversion costs; prices, fees and rounding affect actual receipts. The reward is deducted before the 85% / 15% net-revenue split.

A successful conversion within FEED can earn one thirty-eighth of gross WBNB output, rounded down. The original caller must hold at least 10,000 FATCAT or prove ownership of a position with positive remaining principal. The bounty is deducted before allocating the remaining net revenue: 85% treasury and 15% project development, operations and marketing. It is claimed separately. Later advancement or purchase failures do not cancel this bounty. Standalone conversion, advancement and procurement earn no bounty. Initial conversion bounds are 0.1–50 QQQB per transaction, no hourly cumulative quota, a one-hour TWAP and 300 bps deviation protection; reward procurement has a separate initial 200 bps bound.

For a 10,000-unit trade-value illustration using one common unit of account, the 400 tax routes 20 to Flap, 57 to the team and 323 to the treasury. If the same illustrated income earns a FEED bounty,

the caller receives 10, the team 55.5 and the treasury 314.5, under the same assumptions.

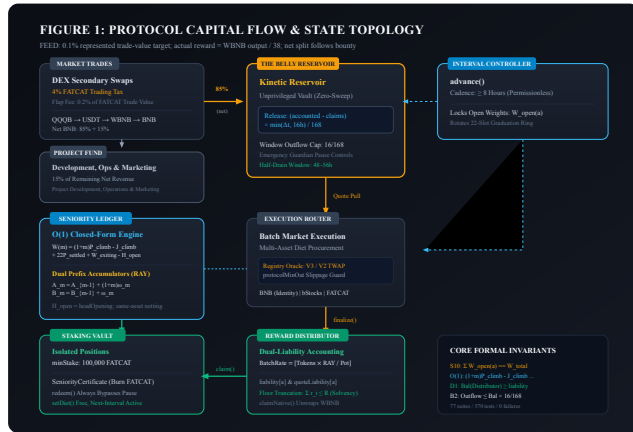


Figure 1: TheFatCat Protocol Capital Flow Topology and State Machine Architecture

2 The Belly Hydrodynamics & Time-Proration

2.1 First-Order Exponential Decay and Steady-State Level

The Belly disburses a fraction of **unreserved quote balance** (accounted - outstandingClaims), not gross contract balance.

At base cadence, intervals require $\Delta t_{\min} = 8$ hours, benchmarked against one week ($T_{\text{week}} = 168$ hours). The nominal release coefficient α is:

$$\alpha = \frac{\Delta t_{\min}}{T_{\text{week}}} = \frac{8}{168} = \frac{1}{21} \approx 4.7619\%$$

This quota uses a week of 168 hours and three meals per day: 1/21 at the standard cadence. Delayed allocation is prorated by seconds but capped at 16 hours. Each successful advance closes one meal and restarts the timer at the current timestamp; it does not backfill every missed meal. A steady-state balance of 21 times per-meal inflow assumes regular eight-hour advances and complete execution.

Let B_k be unreserved quote balance before interval k . Assuming zero new inflow, regular 8-hour ticks, full execution, and negligible dust:

$$B_{k+1} = B_k \cdot (1 - \alpha) = B_k \cdot \left(\frac{20}{21}\right)$$

By mathematical induction:

$$B(k) = B_0 \cdot \left(\frac{20}{21}\right)^k$$

With a constant fee inflow of I every 8 hours, steady-state balance satisfies $B_{\text{eq}} \cdot \alpha = I$:

$$B_{\text{eq}} = \frac{I}{\alpha} = 21 \cdot I$$

Ideal Steady State: Under identical assumptions, steady-state balance is $21I$, equivalent to seven days of continuous inflows. Actual balance varies with batch arrivals, keeper delays, liabilities, and rounding.

2.2 Half-Life Analytical Derivation

Setting $B(k) = \frac{1}{2}B_0$, the half-life interval count $k_{\frac{1}{2}}$ satisfies:

$$\left(\frac{20}{21}\right)^{k_{\frac{1}{2}}} = \frac{1}{2} \implies k_{\frac{1}{2}} = \frac{\ln(0.5)}{\ln\left(\frac{20}{21}\right)} \approx 14.207 \text{ intervals}$$

Converting to physical elapsed time $t_{\frac{1}{2}}$:

$$t_{\frac{1}{2}} = k_{\frac{1}{2}} \times 8 \text{ hours} \approx 113.654 \text{ hours} \approx 4.736 \text{ days}$$

Elapsed Time	Interval k	Remaining %	Exact Value
Day 1 (24h)	3	$\left(\frac{20}{21}\right)^3$	86.38%
Day 3 (72h)	9	$\left(\frac{20}{21}\right)^9$	64.46%
Day 7 (168h)	21	$\left(\frac{20}{21}\right)^{21}$	35.89%
Day 14 (336h)	42	$\left(\frac{20}{21}\right)^{42}$	12.88%
Day 30 (720h)	90	$\left(\frac{20}{21}\right)^{90}$	1.24%

Reserve capital is not mathematically exhausted after 7 days of zero volume; after 30 days, approximately 1.24% of the initial balance remains.

2.3 Dynamic Time-Proration and Discrete Rate Limits

1. Single-Release Span Ceiling (CAP_SPAN):

In `IntervalController.sol`, maximum prorated time is capped at $CAP_SPAN = 2 \times MIN_INTERVAL = 16$ hours:

$$\text{Allocation} = \left\lfloor (\text{accounted} - \text{claims}) \cdot \frac{\min(\Delta t, \text{CAP_SPAN})}{T_{\text{week}}} \right\rfloor$$

2. Vault Outflow Rate Limit (windowAllowance):

In `Belly.sol`, immutable parameters restrict maximum window outflow:

$$\text{MaxOutflow} = \left\lfloor \text{Balance} \cdot \frac{16}{168} \right\rfloor \approx 9.5238\%$$

Even under compromised spenders, continuous 50% drawdown equivalent is:

$$n_{\frac{1}{2}}^{\text{eq}} = \frac{\ln(0.5)}{\ln(\frac{152}{168})} \approx 6.92569 \text{ windows} \approx 55.4055 \text{ hours}$$

Discrete execution requires seven full window allowances (49.6295% remaining after 7th). Spanning 6 intervals, it takes 48 hours if initial allowance is usable, and 56 hours if waiting for window rollover.

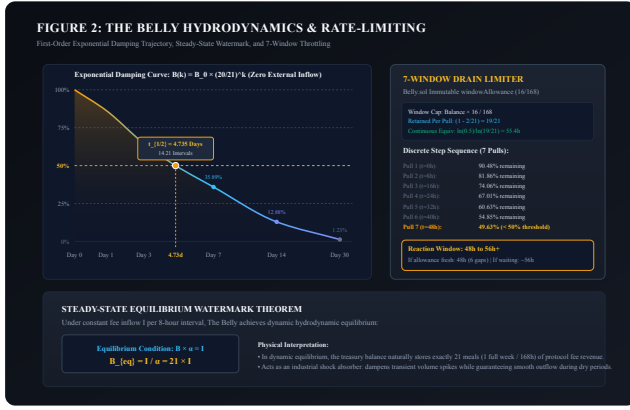


Figure 2: The Belly Hydrodynamic Decay Curve and Discrete Rate-Limiter Defensive Model

3 Seniority Ledger & Scalable Topology

3.1 Independent Positions and State Isolation

Each FATCAT stake creates an isolated position across two contracts:

A position records owner, principal, diet, activeFrom, effectiveFrom and endInterval.

- **Anti-merging:** Positions cannot be merged, preventing seniority whitewashing;
- **Next-interval Activation:** Stakes in interval m become active at activeFrom = $m + 1$, neutralizing flash-loan dividend exploits;
- **Effective Activation Mapping:** For standard positions, effectiveFrom = activeFrom; for positions opened with a Seniority Certificate (via stakeWithCertificate), let certificate head-start be head = multiplier - 1, setting effectiveFrom = activeFrom - head.

3.2 $O(1)$ Closed-Form Seniority Aggregation

Position i with principal p_i accrues seniority notch $c_{i(m)}$. Prior to activation, coefficient is 0; once active, it scales from virtual activation interval effectiveFrom _{i} up to 22:

Let a_i denote actual activation, e_i the weight reference and d_i the exit boundary:

$$c_{i(m)} = \begin{cases} 0 & \text{if } m < a_i \text{ or } m \geq d_i \\ \min(1 + m - e_i, 22) & \text{if } a_i \leq m < d_i \end{cases}$$

Positions divide into:

1. **Climbing Cohort:** $m - \text{effectiveFrom}_i < 21$;
2. **Settled Cohort:** $m - \text{effectiveFrom}_i \geq 21$.

Summing over climbing positions:

$$\sum_{i \in \text{Climb}} p_i (1 + m - e_i) = (1 + m)P_{\text{climb}} - J_{\text{climb}}$$

The contract maintains four global scalars:

- $P_{\text{climb}} = \sum_{i \in \text{Climb}} p_i$
- $J_{\text{climb}} = \sum_{i \in \text{Climb}} (p_i \cdot \text{effectiveFrom}_i)$
- $P_{\text{settled}} = \sum_{i \in \text{Settled}} p_i$
- $\text{headOpening} = \sum_{i \in \text{Opening}} (\text{head}_i \cdot p_i)$ (Opening interval head-start weight)

Total active weight evaluation is strictly $O(1)$, subtracting transient headOpening in the opening interval:

$$W(m) = (1 + m)P_{\text{climb}} - J_{\text{climb}} + 22P_{\text{settled}} + W_{\text{exiting}} - H_{\text{opening}}$$

H_{opening} denotes headOpening.

Each transition processes one global graduation slot and one per active asset. Cost is independent of staker count and scales with the bounded active asset set; archived claim snapshots remain outside per-meal loops.

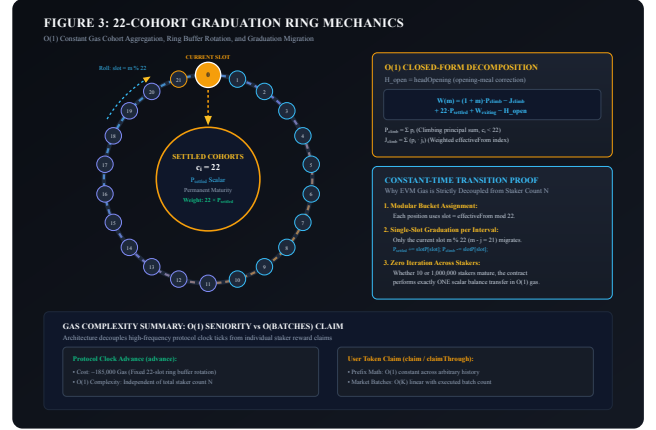


Figure 3: 22-Cohort Graduation Ring Mechanics

3.3 Dual-Accumulator Formulation

Per-interval yield per unit weight for asset a scaled by RAY = 10^{27} :

$$\omega_{a,m} = \left\lfloor \frac{\text{Share}_{a,m} \cdot \text{RAY}}{W_{\text{open}}(a, m)} \right\rfloor$$

Prefix accumulators:

$$A_{a,m} = A_{a,m-1} + (1 + m) \cdot \omega_{a,m}$$

$$B_{a,m} = B_{a,m-1} + \omega_{a,m}$$

Climbing position quote entitlement, using $j = e_i$:

$$\text{Entitlement}(j, m) = \left\lfloor \frac{p \cdot (\Delta A_a - j \cdot \Delta B_a)}{\text{RAY}} \right\rfloor$$

Settled position quote entitlement:

$$\text{Entitlement}_{\text{settled}} = \left\lfloor \frac{22 \cdot p \cdot \Delta B_a}{\text{RAY}} \right\rfloor$$

Dual prefix subtraction yields $O(1)$ entitlement arithmetic per clean interval. Complete token redemption requires $O(K)$ batch iteration, boundable via claimThrough.

3.4 Mid-Interval Forfeiture Netting

Positions can redeem principal without lookup penalties (redeem()), forfeiting the active interval's meal:

$$\text{Forfeited}_a = \left\lfloor \text{Allocation}_a \cdot \frac{w_{\text{exit}}}{W_{\text{open}}(a)} \right\rfloor$$

Unexecuted quote stays in Belly. Forfeits reduce the claimable denominator for remaining claimants of the same asset and funded batch; they do not generally roll into the next meal. Only a disabled asset's provably fully forfeited unexecuted range can be voided and its reservation released.

4 Allocation Neutrality & Seniority Scenarios

4.1 Nominal Quote Allocation Under Ideal Conditions

Under security property S10, $\sum_{a \in \text{MENU}} W_{\text{open}}(a) = W_{\text{total}} \cdot \text{Asset } a \text{ allocation is:}$

$$Q_a = Q_{\text{total}} \cdot \frac{W_{\text{open}}(a)}{W_{\text{total}}}$$

Position i 's entitlement is:

$$\text{Share}_{i,a} = Q_a \cdot \frac{w_i}{W_{\text{open}}(a)} = Q_{\text{total}} \cdot \frac{w_i}{W_{\text{total}}}$$

Denominators cancel in real numbers without probation caps. Position quote budget is $\frac{w_i}{W_{\text{total}}}$, independent of cohort size.

4.2 Seniority Relative Advantage Analysis

Comparison	Multiplier	Economic Context	Context
Max Mature vs Notch 1	22.0x	Temporary disparity, parity in 21 ticks	
Max Mature vs Pool Average	$\frac{22}{\bar{c}}$	Distribution-dependent, $\bar{c} \in [1, 22]$	
Mature in Stagnant Pool	1.0x	All mature, premium converges to 1	

With principal-weighted average seniority $\bar{c} \in [1, 22]$, relative advantage $\frac{22}{\bar{c}}$ spans 1.0x to 22.0x. Uniform distribution ($\bar{c} = 11.5$) yields 1.913x as a representative scenario, not a mathematical upper bound.

Ordinary positions reach 22 after 21 completed active intervals, at least seven days from activation. More principal cannot accelerate the ramp, but a sufficiently large new position can outweigh a smaller mature position.

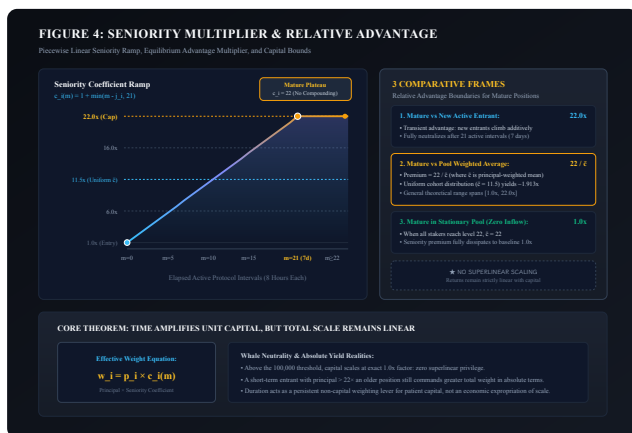


Figure 4: Seniority Multiplier, Settled Premium Curve, and Linear Principal Bounds

4.3 Minimum Stake and Seniority Certificate Binding Architecture

- Immutable Minimum Stake:** minStake set to 100_000 FATCAT (0.01% supply), no setter post-deployment;
- Strict Linear Scaling:** Weight scales strictly linearly with capital beyond threshold;
- Seniority Physicalization Concept & Post-Deployment Binding:**
 - Decoupled Standalone Subsystem:** The Time-Seniority Certificate is an independent on-chain credential system physicalizing accumulated staking duration, decoupled from genesis contracts and designated for subsequent governance activation;
 - Two-Phase Timelock Binding Pipeline:** StakingVault reserves an immutable one-time binding interface (proposeCertificate and activateCertificate) protected by timelock delay; at genesis, the hook is unset (address(0));
 - Mechanism Scope & Separate Disclosure:** Upon future activation, it enables seniority encapsulation, floor exemptions, and supply contraction. Final standards, supply caps, royalties, and release schedules will be published via standalone governance releases.

5 Execution & Solvency Accounting

Procurement and funded fallback settle one whole pending meal for one asset per call. The earliest positive unsettled interval across all listed assets, including disabled assets with pending funds, must clear before later meals spend. Assets in the same earliest meal may settle in any order. A blocked asset can delay other diets' later meals until execution or eligible fallback clears it; clock advancement remains available.

5.1 Dual-Liability Accounting System

RewardDistributor maintains two separate liability ledgers:

- Token Liabilities (liability[asset]):** D18{tok} dimension;
- Quote Liabilities (quoteLiability[asset]):** D18{quote} dimension.

Batch exchange rate:

$$\text{BatchRate} = \left\lfloor \frac{\text{TokensReceived} \cdot \text{RAY}}{\text{ClaimablePot}} \right\rfloor$$

An asset pot pending for more than 24 hours (pendingAge > maxPendingAge, default 86400 seconds) enables permissionless fallbackFinalize(). The clock starts when the pot becomes nonempty; retries and further allocations do not reset it until the pot is emptied. Fallback records WBNB liabilities that users may claim with unwrapping to BNB.

Disabled assets and a first whole meal exceeding the per-swap cap (RewardDistributor.spendFor(asset, maxExecQuote) == 0) qualify for fallback without waiting for timeout. Waiting behind an earlier global meal alone does not grant eligibility. Settlement still requires Belly allowance, global chronology, whole-meal boundaries and a successful transaction. Ordinary claims of settled liabilities do not consume Belly allowance.

5.2 On-Chain TWAP and Execution Bounds

- Initial Menu:** BNB/WBNB, QQQB, SPCXB, NVDAB, SPYB and GOOGLB. WBNB settles directly; the five bStocks use V3 two-hop routes;

2. **Per-Asset Oracles:** bStocks use PancakeV3TwoHopTwapOracle; compatible V2 routes use V2 TWAP. The FATCAT/QQQB launch pair is not a direct FATCAT/WBNB reward quote;
3. **Slippage Protection:** Router verifies:

$$\text{minOut} = \left\lfloor \frac{\text{expectedOut} \cdot (10000 - \text{deviationBps})}{10000} \right\rfloor$$

5.3 Conservative Rounding and Solvency Invariants

For each batch with $\sum_i x_i \leq S$:

$$\text{Rate} = \left\lfloor \frac{R \cdot \text{RAY}}{S} \right\rfloor, \quad r_i = \left\lfloor \frac{x_i \cdot \text{Rate}}{\text{RAY}} \right\rfloor$$

Guarantees:

$$\sum_i r_i \leq R$$

Tested property D1 invariants:

- `balanceOf(Distributor, a) ≥ liability[a]`
- WBNB balance must cover both books together:

$$B_{\text{WBNB}} \geq L_{\text{WBNB}} + \sum_a Q_a$$

Here L denotes liability and Q denotes quoteLiability.

5.4 FEED and Sustainable Menus

FEED uses `ProtocolExecutor.maintain(...)` or the Flap Vault facade to check conversion, one due meal and the cursor-assigned procurement batch. `previewAssets()` determines the exact batch; callers cannot choose a subset. Default menu capacity is 16, with 4 assets checked per FEED; hard limits are 32 and 8, and batch size cannot exceed capacity. Governor queues configuration changes and executes after three days; Governor or Guardian may cancel. Website, Flap and keeper read the same on-chain settings.

A scan does not guarantee a purchase. A successful FEED conversion earns the eligible caller's bounty even if later advancement, reads or purchases fail. Ordinary leg failures are recorded and remaining assets are checked; failed purchases leave budgets pending for retry. Follow-up gas is isolated. If that whole frame fails, only its work rolls back and the cursor stays unchanged. Invalid batches, positive conversion requests with no completed leg, or whole-transaction out-of-gas still revert everything. Included transactions cost caller gas; standalone maintenance remains available.

Disabled assets occupy capacity until weights, pending quote and unconsumed forfeits clear and Governor archives them. Historical claims and final snapshots remain; archived assets leave active settlement loops. WBNB cannot be disabled or removed. FATCAT is bound at Registry deployment and, once listed after graduation and route review, cannot be disabled or removed. This guarantees menu membership, not uninterrupted purchases; failed execution can settle in WBNB. Permanently migrated or archived token addresses cannot be relisted.

6 Security Posture & Dependencies

Staking requires an explicit `openStaking()` transaction: Governor-only before graduation, permissionless afterward. Success starts the seven-day reward clock. `advance()` syncs Belly balances and records budgets; only execution or fallback releases funds.

6.1 Restricted Vault Privileges

1. **The Belly:** Has immutable governor and guardian roles (governance 2-of-3 Gnosis Safe). 15% of remaining net revenue is allocated atomically as project development, operations and marketing income with zero protocol privileges. No admin sweep, rescue, or withdrawal. Spender activation is one-time with timelock;
2. **StakingVault:** `redeem()` intentionally ignores pause. Pause blocks only new deposits.

6.2 Tax Processor External Ownership

Tax processor is retained by Flap factory post-graduation. Tax lasts for 100 years post-graduation (becoming permanently zero thereafter), and after a 1-year anti-farmer period taxes only the primary liquidity pool. Protocol provides `watch-processor.sh` to poll state. Direct custody risk affects future fees, not principal.

6.3 Default Diet and RWA Risk Isolation

1. **WBNB Default and Fallback:** Identity route requires no market swap, eliminating bStock RWA risk;
2. **Probation:** New assets face a 5% cap; initial menu exempt;
3. **FATCAT Listing:** Requires 3-day governance timelock post-graduation.

7 Verification & Disclaimers

7.1 Reproducible Test Evidence

Regression and stateful property tests cover protocol flows and accounting boundaries. Verification records identify the source revision, command, results and skipped checks. Property tests do not equate to formal verification.

7.2 Deterministic Scope and Non-Goals

1. **Not a Fixed-Income Product:** Zero APY commitments;
2. **Not a Lockup:** Principal withdrawable in any block;
3. **Zero Superlinear Bias:** Capital scales strictly linearly;
4. **Not a Pension:** Half-life 4.736 days under zero volume; 1.24% balance after 30 days.

8 Appendix A: Notation and Units

Symbol	Semantic Definition	Domain Bound	/	EVM Dimension Scale
m	Global interval index	$m \geq 0$		Dimensionless (uint256)
a_i	Actual activation	$m_{\text{stake}} + 1$		Interval index
$e_i = j_i$	Weight reference	$0 \leq e_i \leq a_i$		Dimensionless
p_i	Staked principal	$\geq 100,000$		D18 {FATCAT}
$c_{i(m)}$	Seniority notch	Integer [0, 22]		Dimensionless scalar
$w_{i(m)}$	Effective weight	$p_i \cdot c_{i(m)}$		D18 {weight}

Symbol	Semantic Definition	Domain Bound	/	EVM Dimension Scale	Di-
$W(m)$	Total active weight	$\sum w_{i(m)}$		D18 {weight}	
W_{open}	Asset open denominator	$\sum W_{\text{open}} = W$		D18 {weight}	
P_{climb}	Climbing principal sum	$\sum p_i$		D18 {FATCAT}	
J_{climb}	Climbing weighted virtual index	$\sum (p_i \cdot e_i)$		D18 {FATCAT}	
P_{settled}	Settled principal sum	$\sum p_i$		D18 {FATCAT}	
H_{opening}	Opening interval head weight	$\sum (\text{head}_i \cdot p_i)$		D18 {weight}	
W_{exiting}	Residual exit weight	$\sum w_i$		D18 {weight}	
RAY	Accumulator scale	Constant 10^{27}		D27 scale	
$\omega_{a,m}$	Yield per unit weight	Yield scalar		D27 (uint256)	
$A_{a,m}$	Accumulator A	$\sum (1 + k)\omega$		D27 prefix sum	
$B_{a,m}$	Accumulator B	$\sum \omega$		D27 prefix sum	
α	Belly release fraction	$\frac{8}{168} = \frac{1}{21}$		Dimensionless ($\approx 4.76\%$)	
B_k	Belly available balance	Unreserved quote		D18 {quote} (WBNB)	
r_b	Batch conversion rate	Fixed exchange rate		D27 ratio	
L_a	Token liability	≥ 0		D18 {tok}	
Q_a	Quote asset liability	≥ 0		D18 {quote}	

9 Appendix B: Solidity Rounding and Solvency Bounds

9.1 Integer Floor Division

$$\forall a, b \in \mathbb{N}^+, \left\lfloor \frac{a}{b} \right\rfloor = \frac{a - (a \bmod b)}{b} \leq \frac{a}{b}$$

Rounding error $\varepsilon = \frac{a \bmod b}{b}$ satisfies $0 \leq \varepsilon < 1$.

9.2 Non-negative Dust Theorem

Theorem: Given token purchase R , pot S , and $\sum_{i=1}^N x_i \leq S$:

$$\sum_{i=1}^N r_i \leq R$$

Proof:

1. Rate definition: $\text{Rate} = \left\lfloor \frac{R \cdot \text{RAY}}{S} \right\rfloor$
2. Position claim: $r_i = \left\lfloor \frac{x_i \cdot \text{Rate}}{\text{RAY}} \right\rfloor \leq \frac{x_i \cdot \text{Rate}}{\text{RAY}}$
3. Summing across N positions:

$$\sum_{i=1}^N r_i \leq \frac{\text{Rate}}{\text{RAY}} \sum_{i=1}^N x_i$$

4. Substituting $\sum x_i \leq S$ and $\text{Rate} \leq \frac{R \cdot \text{RAY}}{S}$:

$$\sum_{i=1}^N r_i \leq \frac{\text{Rate} \cdot S}{\text{RAY}} \leq \frac{\left(\frac{R \cdot \text{RAY}}{S}\right) \cdot S}{\text{RAY}} = R \quad \blacksquare$$

Corollary: Residual dust $\Delta_{\text{dust}} = R - \sum_{i=1}^N r_i \geq 0$ permanently settles in the distributor, ensuring physical balance dominates recorded liabilities (Balance $\geq$ Liability).

9.3 Solvency Preconditions

1. **Standard Token Behavior:** No fee-on-transfer, no negative rebate;
2. **Liability Monotonicity:** `finalize()` and `fallbackFinalize()` increment atomically; `claim()` decrements atomically;
3. **Ownerless Distributor:** No administrative sweep or drain functions.

10 Appendix C: Transaction Cost and Simulation

Gas depends on contract state, routes and batch history. Simulate the exact transaction; the Through interfaces provide caller-selected cutoffs, not a fixed gas ceiling.

Operation	Complexity	Cost Drivers
<code>stake()</code>	$O(1)$	Token transfer and a new position
<code>stakeWithCertificate()</code>	$O(1)$	Position and certificate lock
<code>redeem()</code>	$O(1)$	Position exit and token transfer
<code>redeemAndIssueCertificate()</code>	$O(1)$	Position exit and certificate mint
<code>setDiet()</code>	$O(1)$	Diet change bookkeeping
<code>advance()</code>	$O(M)$	Listed asset count M
<code>claim()</code>	$O(K)$	Covered batch count K
<code>claimThrough()</code>	$O(K)$	Batches through the selected cutoff
<code>claimNative()</code>	$O(K)$	Covered batches and WBNB unwrap

11 Appendix D: Contract Components and Verification Boundaries

Component	Network	Verification Role
FATCAT Token	BNB Chain	1B fixed supply, immutable clone
PancakeSwap V2 Pool	BNB Chain	Locked liquidity, TWAP price feed

Component	Network	Verification Role
FatCatQqqbVault	BNB Chain	QQQB $\rightarrow$ USDT $\rightarrow$ WBNB $\rightarrow$ BNB; Beacon proxy
FatCatQqqbVault Factory	BNB Chain	Factory owns Beacon; Flap Guardian alone may upgrade or lock
FatCatStaking Vault	BNB Chain	Fixed BNB receiver; net: 85% Belly treasury, 15% team Ops
The Belly Vault	BNB Chain	Unprivileged exponential damped reservoir
StakingVault	BNB Chain	Isolated staking, unpausable redeem
SeniorityLedger	BNB Chain	$O(1)$ closed-form algebraic ledger
RewardAsset Registry	BNB Chain	MENU registry with 5% probation cap
ExecutionRouter	BNB Chain	Delayed activation, MEV-protected router
RewardDistributor	BNB Chain	Dual-liability independent accounting
Seniority Certificate (ERC-721)	BNB Chain	Standalone post-deployment subsystem; decoupled from genesis vaults, bound via time-locked proposal
PancakeV2 TwapOracle	BNB Chain	Endogenous TWAP price reader

12 Appendix E: Responsible Disclosure & Security Channel

1. **Official Security Channel:** Submit reports via GitHub Private Security Advisories;
2. **Review & Feedback:** Regular review with prompt triage feedback upon technical evaluation;
3. **White-Hat Grants:** Critical and High vulnerabilities receive discretionary grants disbursed from Ops Safe operational funds. The reward core and BNB receiver have fixed implementations. Flap Guardian manages upstream QQQB vault upgrades through the Factory; core replacements require migration procedures.

— Mathematics Belong to Everyone, Code is Law —